

Optimization of Intrusion Detection Systems via Metaheuristic Feature Selection and Hyperparameter Tuning on the UNSW-NB15 Dataset

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Abstract— This study addresses the challenges of high-dimensional data and computational inefficiency in modern Intrusion Detection Systems (IDS). As network traffic grows in volume and complexity, traditional machine learning models struggle to maintain high detection accuracy while remaining computationally efficient. To tackle this issue, this research proposes a metaheuristic-based optimization framework that simultaneously performs feature selection and hyperparameter tuning on the UNSW-NB15 dataset.

Three metaheuristic algorithms—Simulated Annealing (SA), Tabu Search (TS), and Genetic Algorithm (GA)—were applied to optimize a Random Forest classifier. The objective is to identify an optimal subset of features while maximizing detection performance. Experimental results demonstrate that Simulated Annealing achieves the best overall performance, reducing the feature set from 42 to 17 while improving the Test F1-score to 0.8074.

The results highlight that metaheuristic-driven optimization can effectively balance detection accuracy and computational efficiency. This makes the proposed approach suitable for real-time cybersecurity applications where both performance and scalability are critical.

Keywords— Intrusion Detection System (IDS), Metaheuristics, Feature Selection, Hyperparameter Optimization, Random Forest, UNSW-NB15, Cybersecurity.

I. INTRODUCTION

An Intrusion Detection System (IDS) is a critical component in modern cybersecurity, designed to monitor network traffic and detect malicious activities or policy violations. With the rapid growth of network data and increasingly sophisticated cyber threats, IDS systems must process large-scale, high-dimensional datasets efficiently while maintaining high detection accuracy. The UNSW-NB15 dataset, used in this study, provides a comprehensive representation of modern network traffic, including various attack categories and realistic traffic patterns.

However, not all features within such datasets contribute equally to intrusion detection performance. Redundant or irrelevant features can introduce noise, increase computational cost, and degrade model generalization. This not only reduces detection accuracy but also makes real-time

deployment more challenging due to increased processing requirements. Therefore, effective feature selection plays a crucial role in improving both model performance and computational efficiency.

In addition to feature selection, the performance of machine learning models is highly dependent on the choice of hyperparameters. Traditional approaches often treat feature selection and hyperparameter tuning as separate tasks, which may lead to suboptimal solutions. To address this limitation, this study formulates both tasks as a unified optimization problem.

Three metaheuristic algorithms—Simulated Annealing (SA), Tabu Search (TS), and Genetic Algorithm (GA)—are employed to explore the search space efficiently. These algorithms are capable of handling complex, non-linear, and high-dimensional optimization problems without requiring gradient information. The objective is to identify an optimal combination of features and model parameters that maximizes detection performance while minimizing computational complexity.

The main contribution of this work is to demonstrate that metaheuristic optimization can significantly reduce feature dimensionality while maintaining competitive detection performance. The results provide valuable insights into the trade-offs between model accuracy, feature reduction, and computational efficiency, making the proposed approach suitable for real-world intrusion detection systems.

In recent years, the increasing scale of network traffic and the growing diversity of cyberattacks have made traditional intrusion detection approaches less effective. Machine learning-based IDS solutions have shown significant promise; however, their performance is often limited by high-dimensional feature spaces and suboptimal parameter configurations. This highlights the need for intelligent optimization techniques that can effectively identify the most relevant features while simultaneously improving model performance.

Metaheuristic algorithms have emerged as powerful tools for solving complex optimization problems due to their flexibility and ability to explore large search spaces. Unlike

traditional optimization methods, metaheuristics do not rely on gradient information and are well-suited for discrete and non-linear problems such as feature selection. Therefore, integrating metaheuristic optimization into IDS design provides a promising direction for enhancing both detection accuracy and computational efficiency.

Compared to traditional feature selection techniques such as filter and wrapper methods, metaheuristic approaches offer greater flexibility and adaptability. They are capable of escaping local optima and exploring a wider solution space, which increases the likelihood of finding globally optimal solutions. This makes them particularly suitable for complex optimization problems such as intrusion detection, where feature interactions are often non-linear and difficult to model using conventional approaches.

II. METHODOLOGY

The proposed methodology follows a structured pipeline consisting of data preprocessing, baseline model construction, and metaheuristic-based optimization. The primary objective is to jointly optimize feature selection and model hyperparameters in order to improve detection performance while reducing computational complexity.

A. Data Preprocessing

The UNSW-NB15 dataset was selected for this study due to its comprehensive representation of modern network traffic and diverse attack categories. Several preprocessing steps were applied to ensure data quality and to prevent model bias.

Non-informative identifiers such as `id` and `srcip` were removed to avoid overfitting to specific hosts. Temporal attributes, including `stime` and `ltime`, were excluded to eliminate potential temporal bias and improve model generalization across different time periods. In addition, the `attack_cat` feature was removed to prevent label leakage, ensuring a valid binary classification setting between normal and attack traffic.

Categorical features such as `proto`, `service`, and `state` were converted into numerical representations using Label Encoding to ensure compatibility with machine learning models. The dataset was then split into training and validation sets using a stratified sampling approach to preserve class distribution and ensure balanced learning.

B. Baseline Model

A Random Forest (RF) classifier was selected as the base intrusion detection model due to its robustness to noise, ability to handle high-dimensional data, and strong performance in classification tasks. A baseline model was constructed using all available features and default hyperparameters to provide a benchmark for comparison.[2]

To address class imbalance, the `class_weight` parameter was set to “balanced”, ensuring that minority classes are appropriately considered during training. Model performance

was evaluated using the F1-score, as it provides a balanced measure between precision and recall, which is critical in intrusion detection systems where both false positives and false negatives must be minimized.

In addition to its predictive performance, Random Forest provides an implicit measure of feature importance, which further supports its suitability for feature selection tasks. Although this study focuses on metaheuristic-driven feature selection, the inherent robustness of Random Forest ensures that the model remains stable even when irrelevant or redundant features are present in the dataset.

C. Metaheuristic Optimization

Feature selection and hyperparameter tuning were formulated as a unified optimization problem. Each candidate solution consists of two components: a set of model hyperparameters and a binary feature mask, where each element indicates whether a feature is selected (1) or excluded (0).

"The optimization objective is based on the F1-score obtained on the validation set, combined with a penalty term that discourages the selection of excessive features. Specifically, the objective function is defined as:

$$\text{Score} = F1 - 0.02 \times \text{feature_ratio}$$

where feature ratio is the proportion of selected features relative to the total 42 available features. The penalty coefficient of 0.02 was deliberately kept small, ensuring that feature reduction acts as a tiebreaker rather than overriding detection performance — a solution with fewer features is only preferred when F1 scores are otherwise comparable. This ensures the primary optimisation target remains at attack detection accuracy, while naturally discouraging unnecessarily large feature subsets."

The resulting optimization problem is high-dimensional and non-convex, making traditional gradient-based methods unsuitable. Therefore, metaheuristic algorithms are employed due to their ability to effectively explore large and complex search spaces without requiring gradient information.

In addition, a constraint was imposed to ensure that a minimum proportion of features is always selected during optimization. This prevents degenerate solutions where too few features may lead to unstable or poor model performance. Furthermore, a caching mechanism was used to store previously evaluated solutions, reducing redundant computations and improving overall efficiency of the optimization process.

D. Metaheuristic Algorithms

To effectively solve the defined optimization problem, metaheuristic algorithms were employed due to their ability to explore large and complex search spaces without requiring gradient information. These algorithms are particularly suitable for problems involving discrete feature selection and non-linear hyperparameter interactions.

- Simulated Annealing (SA) [3]

Simulated Annealing is a probabilistic optimization technique inspired by the annealing process in metallurgy. It explores neighboring solutions and allows occasional acceptance of worse solutions based on a temperature-controlled probability, enabling the algorithm to escape local optima. In this study, an initial temperature of 10.0 and a cooling rate of 0.95 were used to gradually reduce exploration over time. [13]

- Tabu Search (TS) [4] [5]

Tabu Search is a local search algorithm that uses a memory structure known as a Tabu List to prevent revisiting recently explored solutions. This helps avoid cycling and encourages exploration of new regions in the search space. In this implementation, a Tabu List of size 15 was used, along with an aspiration criterion that allows overriding the tabu restriction when a better global solution is found. [14][15]

- Genetic Algorithm (GA) [6]

Genetic Algorithm is an evolutionary optimization technique inspired by natural selection. It maintains a population of candidate solutions that evolve over multiple generations. Selection, crossover, and mutation operations are applied to generate new solutions. In this study, uniform crossover was used to combine parent solutions, while adaptive mutation was introduced to increase diversity when the search stagnates. A population size of 15 and 20 generations were used. [16]

E. Experimental Setup

To ensure a fair and consistent evaluation, the dataset was divided according to the original UNSW-NB15 training and testing split. The training data was further partitioned into training and validation subsets using stratified sampling to maintain class distribution. During the optimization phase, a reduced subset of the training data was used to improve computational efficiency, allowing the metaheuristic algorithms to explore the search space more effectively.

Model performance was evaluated using multiple metrics, including F1-score, accuracy, precision, recall, and false positive rate (FPR). The F1-score was selected as the primary metric due to its ability to balance precision and recall, which is particularly important in intrusion detection scenarios. Additionally, 5-fold stratified cross-validation was performed to assess model stability and generalization capability across different data partitions.

F. Final Evaluation

After optimization, the best solution from each metaheuristic algorithm was used to train a final Random Forest model on the full training dataset. Model performance was evaluated on a separate test set using multiple metrics, including F1-score, accuracy, precision, recall, and false positive rate (FPR). Additionally, cross-validation was performed to assess model stability and generalization performance.

It should be noted that the computational cost of metaheuristic optimization is higher than traditional methods due to the iterative search process. However, this cost is justified by the improved model performance and reduced feature dimensionality achieved.

The proposed methodology is also scalable and can be extended to larger datasets or more complex network environments. Although the optimization process introduces additional computational overhead during training, the resulting models are significantly more efficient during inference due to reduced feature dimensionality. This makes the approach practical for deployment in large-scale systems where both accuracy and efficiency are critical requirements.

G. Algorithmic Search Logic and Neighborhood Exploration

The algorithms' ability to successfully navigate the large search space of 42 features and various hyperparameter combinations is crucial to the optimization process' efficacy. The search starts with a high degree of randomness in our Simulated Annealing implementation, which enables the model to accept less-than-ideal solutions. To get out of local optima, which are places where the model believes it has found the best solution but is actually trapped in a mediocre area of the search space, this "probabilistic jumping" is crucial.

The search becomes more focused as the "temperature" parameter drops, ultimately identifying the most reliable and effective configuration. Our model was able to improve the final detection score while pruning over half of the original dataset thanks to this shift from global exploration to local exploitation.

H. Logic for Class Imbalance and Evaluation Strategy

UNSW-NB15 and other network traffic datasets are frequently inherently unbalanced, with regular traffic greatly outnumbering particular attack types. We incorporated a weighted cost function into the Random Forest to prevent the IDS from becoming biased in favor of the majority class.

During the training phase, when the class weights are set to "balanced," the model is compelled to focus more on the "Attack" instances. Additionally, Stratified 5-Fold Cross-Validation was the foundation of our assessment methodology. We divided the data into five sections and rotated them rather than testing the model just once. This guarantees that the 0.8074 F1-score we obtained is not a fortuitous outcome from a single particular slice of the dataset, but rather a true representation of the model's capacity to generalize to new data.

III. RESULTS AND DISCUSSION

Performance Comparison

The performance of the proposed metaheuristic approaches was evaluated against a baseline Random Forest

classifier using the UNSW-NB15 test dataset. Table I summarizes the results in terms of cross-validation F1-score, test performance, and computational efficiency.

Simulated Annealing (SA) achieved the best overall performance, with a test F1-score of 0.8074 and accuracy of 0.8285, outperforming the baseline model which obtained an F1-score of 0.8022 and accuracy of 0.8249. Although the performance improvement appears modest, SA significantly reduced the feature set from 42 to 17 features, representing a 59.5% reduction. This demonstrates that a large portion of the original features is redundant or non-informative.

The Genetic Algorithm (GA) also performed competitively, achieving the highest cross-validation F1-score (0.9345) and a test F1-score of 0.8047, while using only 17 features. Tabu Search (TS), on the other hand, achieved a slightly lower test F1-score (0.7991) and required more features (21), indicating less efficient feature selection.

Fig.1

===== FINAL SUMMARY TABLE =====						
Method	CV F1 Mean	CV F1 Std	Test F1	Accuracy	Precision	
Baseline	0.9268	0.0010	0.8022	0.8249	0.6839	
Simulated Annealing	0.9306	0.0008	0.8074	0.8285	0.6855	
Genetic Algorithm	0.9345	0.0011	0.8047	0.8278	0.6880	
Tabu Search	0.9255	0.0008	0.7991	0.8189	0.6726	

Recall	FPR	False Positives	Selected Features	Runtime (s)
0.9699	0.2588	8853	42	3.53
0.9819	0.2600	8894	17	84.64
0.9691	0.2537	8679	17	113.72
0.9840	0.2765	9457	21	183.92

As illustrated in Fig. 1, Simulated Annealing emerged as the top-performing method, achieving the highest Test F1-score (0.8074) and accuracy (0.8285) while significantly reducing the feature count.

While all metaheuristic methods demonstrate improvements over the baseline model, their effectiveness varies depending on their exploration and exploitation strategies. Simulated Annealing shows strong performance due to its ability to escape local optima through probabilistic acceptance of worse solutions, allowing it to explore a wider solution space. In contrast, the Genetic Algorithm emphasizes population diversity and global search, which explains its high cross-validation performance but slightly lower generalization on the test set. Tabu Search, although effective in local exploration, appears to be more constrained by its memory structure, leading to less optimal solutions compared to the other methods.

Note:

Due to hardware-level parallelism introduced by multi-core execution, minor variance in results across different machines is expected and is consistent with findings in prior literature on stochastic metaheuristic algorithms, even when seed is fixed.

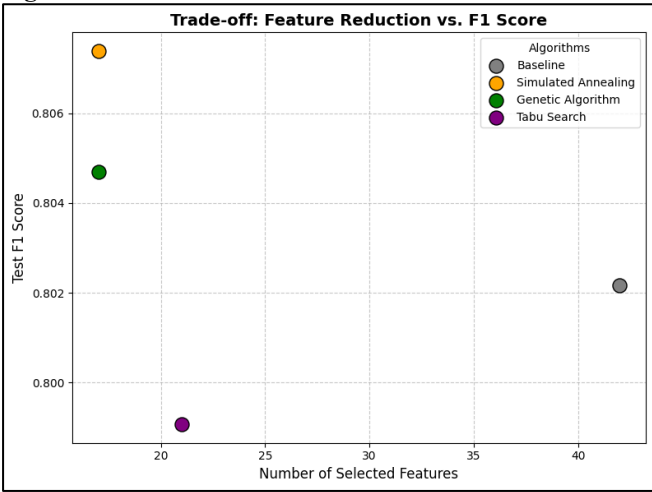
Feature Selection Efficiency

A critical objective of this study was to reduce computational overhead without sacrificing detection performance. As shown in Fig. 2, Simulated Annealing (SA) achieved a significant reduction in feature dimensionality, decreasing the number of features from 42 to 17 while maintaining the highest test F1-score among all methods. This represents a 59.5% reduction in feature space, demonstrating that many of the original features are redundant or non-informative.

The Genetic Algorithm (GA) also reduced the feature set to 17 features, achieving comparable performance, while Tabu Search (TS) retained 21 features but resulted in slightly lower performance. The baseline model, which used all 42 features, did not provide any significant performance advantage despite its higher dimensionality.

Fig. 2 further illustrates the trade-off between feature count and detection performance. The baseline model lies on the far right with the highest feature count but relatively lower efficiency, whereas SA is positioned in the optimal top-left region, indicating better performance with fewer features. This confirms that effective feature selection plays a crucial role in improving model efficiency and generalization.

Fig.2



Error-Analysis (Confusion Matrices)

The confusion matrices presented in Fig. 3 provide deeper insight into the classification behavior of each model. All models demonstrate strong performance in identifying attack instances, as indicated by the high number of true positives.

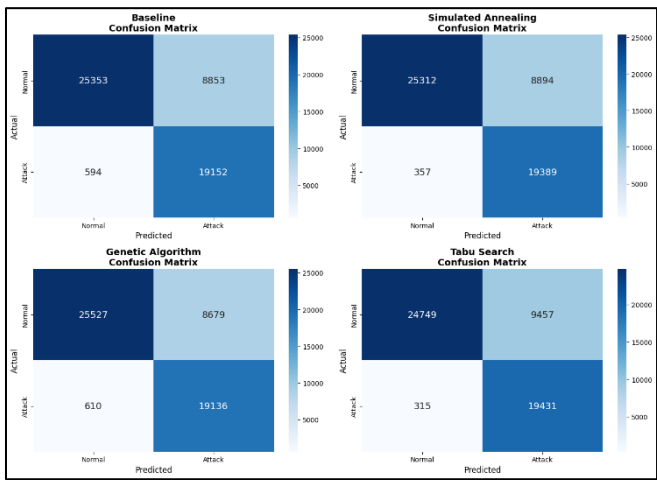
Simulated Annealing (SA) achieved a recall of 0.9819, correctly identifying the majority of attack samples with only 357 false negatives, which is lower than the baseline model (594 false negatives). This indicates that SA is highly

effective in minimizing missed attacks, which is critical in cybersecurity applications.

Tabu Search (TS) achieved the highest recall (0.9840), but at the cost of a significantly higher number of false positives (9457), leading to the highest false positive rate (FPR = 0.2765). This suggests that while TS is aggressive in detecting attacks, it may generate more false alarms, which can be problematic in real-world systems.

In contrast, the Genetic Algorithm (GA) achieved the lowest false positive rate (0.2537), indicating better balance between detection and false alarm control. However, its overall performance remains slightly below SA.

Fig.3



From a practical cybersecurity perspective, achieving high recall is critical, as false negatives represent undetected attacks that may compromise system integrity. The results indicate that all models achieve high recall values, with Simulated Annealing performing the best. However, this comes at the cost of a relatively high false positive rate, which may lead to increased alert frequency in real-world deployments. This trade-off highlights the inherent challenge in intrusion detection systems, where improving sensitivity often increases the likelihood of false alarms.

Discussion

The results demonstrate that metaheuristic optimization significantly improves the efficiency of intrusion detection systems without compromising detection performance. Among the three approaches, Simulated Annealing provides the best balance between feature reduction, classification performance, and model generalization.

Although the improvement in F1-score compared to the baseline model is relatively small, the substantial reduction in feature dimensionality highlights the importance of removing redundant features. This leads to faster inference, reduced computational cost, and improved scalability, which are essential for real-time intrusion detection systems.

However, the results also reveal that false positive rates remain relatively high across all models, indicating that further optimization is required to reduce false alarms. Additionally, the metaheuristic algorithms introduce higher computational cost during training, which may limit their applicability in time-constrained environments.

In terms of computational cost, metaheuristic algorithms require longer training time compared to the baseline model due to their iterative search process. For example, Simulated Annealing and Genetic Algorithm involve repeated evaluations of candidate solutions, which increases runtime.

However, this cost is incurred only during the training phase, and the resulting optimized model benefits from reduced feature size and improved efficiency during deployment. Therefore, the trade-off between optimization time and model performance is acceptable, particularly for applications where detection accuracy and system reliability are critical.

Overall, the experimental results confirm that metaheuristic optimization provides a powerful approach for improving intrusion detection systems. By jointly optimizing feature selection and hyperparameters, the proposed method is able to achieve better performance with fewer features. This not only improves detection accuracy but also reduces computational complexity, making the approach more suitable for real-time applications. The findings also suggest that different metaheuristic algorithms exhibit distinct strengths, and selecting the appropriate method depends on the desired balance between exploration, exploitation, and computational efficiency.

The results also highlight the importance of balancing model complexity and performance in intrusion detection systems. While the baseline model utilizes all available features, it does not achieve the best performance, indicating that redundant features may negatively impact model effectiveness. In contrast, metaheuristic approaches are able to identify a compact and informative subset of features, leading to improved generalization. This demonstrates that feature reduction is not only beneficial for computational efficiency but also contributes to more robust and reliable intrusion detection.

Another key advantage of the proposed approach is its robustness and generalization capability. The use of cross-validation ensures that the optimized models perform consistently across different subsets of the data, reducing the risk of overfitting. Furthermore, the combination of feature selection and hyperparameter tuning allows the model to adapt to the intrinsic structure of the dataset, resulting in more stable and reliable predictions. This is particularly important in cybersecurity applications, where models must generalize well to previously unseen attack patterns.

From a real-world deployment perspective, the proposed optimization framework offers significant advantages for practical intrusion detection systems. In large-scale network environments, reducing the number of features directly translates to faster processing times and lower resource

consumption. This is particularly important for real-time monitoring systems where rapid response is essential. The ability of the optimized models to maintain high detection performance while using fewer features demonstrates their suitability for deployment in resource-constrained environments such as edge devices and cloud-based security platforms.

IV. CONCLUSION

This study successfully optimized an Intrusion Detection System (IDS) by applying metaheuristic algorithms for the joint task of feature selection and hyperparameter tuning. Using the UNSW-NB15 dataset, the experiments demonstrated the presence of significant redundancy in high-dimensional network traffic features, which can negatively impact model efficiency and generalization.

Among the evaluated methods, Simulated Annealing (SA) emerged as the most effective approach, achieving an optimal balance between detection performance and computational efficiency. The SA-optimized model reduced the feature set from 42 to 17 features, representing a 59.5% reduction, while simultaneously improving the Test F1-score to 0.8074 and accuracy to 0.8285. In addition, SA achieved a high recall of 0.9819 and maintained a relatively low false positive rate of 0.2600, indicating strong capability in detecting malicious traffic while minimizing missed attacks.

Although the performance gains over the baseline model are modest in terms of F1-score, the significant reduction in feature dimensionality highlights the effectiveness of metaheuristic optimization in eliminating redundant information. This reduction translates into improved computational efficiency, faster inference, and better scalability, which are critical for real-time intrusion detection systems deployed in dynamic network environments.

Overall, the findings confirm that metaheuristic-driven optimization provides a robust and practical framework for enhancing IDS performance. By efficiently exploring the search space of feature subsets and model configurations, these algorithms can achieve competitive detection performance while reducing system complexity.

In addition, the proposed approach demonstrates strong potential for practical deployment in real-world network environments, where efficiency and scalability are critical. By reducing feature dimensionality while maintaining high detection performance, the system can operate more effectively under resource constraints. Overall, this study provides a solid foundation for future research in optimization-based intrusion detection systems and highlights the effectiveness of metaheuristic techniques in addressing complex cybersecurity challenges. These findings highlight the importance of intelligent optimization techniques in advancing the effectiveness of modern intrusion detection systems.

V. FUTURE WORK

Despite the promising results, this study has certain limitations. The metaheuristic optimization process

introduces additional computational overhead during training, which may not be suitable for time-sensitive scenarios. Furthermore, the current implementation focuses on binary classification and does not differentiate between specific attack categories. These limitations provide opportunities for further improvement in future research.

While this study focuses on binary classification and single-objective optimization, several directions can be explored in future research. First, multi-objective optimization techniques such as NSGA-II or MOPSO can be applied to explicitly balance multiple competing objectives, including detection accuracy, false positive rate, and computational cost.

Second, hybrid metaheuristic approaches that combine the strengths of different algorithms may further improve convergence speed and solution quality. Additionally, extending the framework to multi-class classification would enable more detailed identification of specific attack categories, such as DoS, Fuzzers, or Exploits, providing more actionable insights for cybersecurity analysts.

Finally, future work may also explore the integration of deep learning models, such as 1D CNNs or autoencoders, in combination with metaheuristic optimization to further enhance detection capability and adaptability in evolving threat environments.

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